

PmTrack: Enabling Personalized mmWave-based Human Tracking

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The difficulty in obtaining targets' identity poses a significant obstacle to the pursuit of personalized and customized millimeter-wave (mmWave) sensing. Existing solutions that learn individual differences from signal features have limitations in practical applications. This paper presents a Personalized mmWave-based human Tracking system, PmTrack, by introducing inertial measurement units (IMUs) as identity indicators. Widely available in portable devices such as smartwatches and smartphones, IMUs utilize existing wireless networks for data uploading of identity and data, and are therefore able to assist in radar target identification in a lightweight manner with little deployment and carrying burden for users. PmTrack innovatively adopts orientation as the matching feature, thus well overcoming the data heterogeneity between radar and IMU while avoiding the effect of cumulative errors. In the implementation of PmTrack, we propose a comprehensive set of optimization methods in detection enhancement, interference suppression, continuity maintenance, and trajectory correction, which successfully solved a series of practical problems caused by the three major challenges of weak reflection, point cloud overlap, and body-bounce ghost in multi-person tracking. In addition, an orientation correction method is proposed to overcome the IMU gimbal lock. Extensive experimental results demonstrate that PmTrack achieves an identification accuracy of 98% and 95% with five people in the hall and meeting room, respectively.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computer systems organization** → **Embedded and cyber-physical systems**.

Additional Key Words and Phrases: mmWave radar, IMU, device-free tracking, human identification

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1 INTRODUCTION

1.1 Motivation & Problem Statement

Through the analysis and utilization of radio frequency (RF) signals, wireless sensing technology enables non-intrusive human sensing in a wide range of applications and scenarios [75]. For example, vital sign detection obtains the physical condition of individuals, which is crucial for medical care and health management; gesture

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recognition can provide convenient and comfortable human-machine interaction services in smart homes. Localization and tracking technologies play a fundamental role in wireless human sensing, as they provide precise position information for better signal processing and analysis [12]. Compared with computer vision-based methods, wireless sensing reduces the risk of privacy leakage and is not affected by visual factors such as lighting and visibility. Among RF technologies, mmWave radar technology using the frequency-modulated continuous wave (FMCW) mechanism possesses high frequency and large bandwidth, and thus has high distance, angle, and displacement resolutions, which enables device-free localization and sensing with unprecedented accuracy [51]. However, as with other device-free sensing technologies, identifying radar targets can be challenging because reflection signals do not directly transmit identity codes like device-based communication, which poses a significant obstacle for highly intelligent and personalized sensing applications.

1.2 Limitations of Prior Art

Most studies learn individual differences from signal features for personalized tracking, such as the gait [39, 42, 45, 72], 3D point cloud [7, 79], and behavior pattern [22]. Although these approaches avoid extra sensors and achieve device-free identification, they still face the following problems that restrict their practical deployment. First, the collection of training data is inevitable, which is time-consuming and infeasible for temporary users such as visitors and conference attendees. Second, the distinctiveness of signal features decreases as the number of people increases. Third, the recognition can be impacted by the diversity and variability of application scenarios, as abstract signal features are sensitive to the environment. Fourth, the timeliness of training data is limited because human behavioral patterns may change over time.

1.3 PmTrack in a Nutshell

To overcome the limitations of existing solutions, this paper presents a Personalized mmWave-based human Tracking system, PmTrack, by introducing IMUs as identity indicators. The adoption of IMUs brings three advantages. First, IMUs have been widely embedded in portable devices such as smartwatches and smartphones [29, 57, 80], therefore users have almost no burden of carrying them. Second, IMUs naturally provide identity during data transmission, with efficient and robust identity feedback. Thirdly, IMU data is uploaded through existing wireless networks such as WiFi and Bluetooth via portable devices, thus avoiding additional implementation costs. As shown in Fig. 1, by correlating data from radar and IMU, lightweight personalized tracking can be enabled, endowing mmWave sensing systems with identity directionality. To associate IMUs with radar targets, we need to address the data heterogeneity. While the radar-measured trajectory can be matched with the IMU computed trajectory, the latter will be severely affected by the cumulative error of the accelerometer [11]. For stable and reliable data matching, PmTrack adopts an intuitive and accurate trajectory description manner—orientation. The PmTrack system essentially recognizes "*who carries which IMU*" through the following three stages.

Radar-based multi-person tracking: after collecting reflection signals, the radar system performs point cloud generation and target detection in the distance-azimuth dimensions. Based on multiple frames of point clouds, we adopt the point target tracking scheme to perform trajectory tracking and management. The orientation of each trajectory is represented by its direction.

IMU attitude tracking: the IMU system performs data fusion on data of the accelerometer and gyroscope to track its attitude in real-time. We choose the yaw in z-y-x rotation order as the orientation of the IMU.

Orientation-based association: PmTrack first performs moving detection on each trajectory to determine the effective intervals of the data, based on which the orientations of trajectory and IMU are extracted for matching. The back projection algorithm is adopted to calculate the data similarity, which are finally fed into the Hungarian algorithm to achieve the matching of trajectory and person identity.

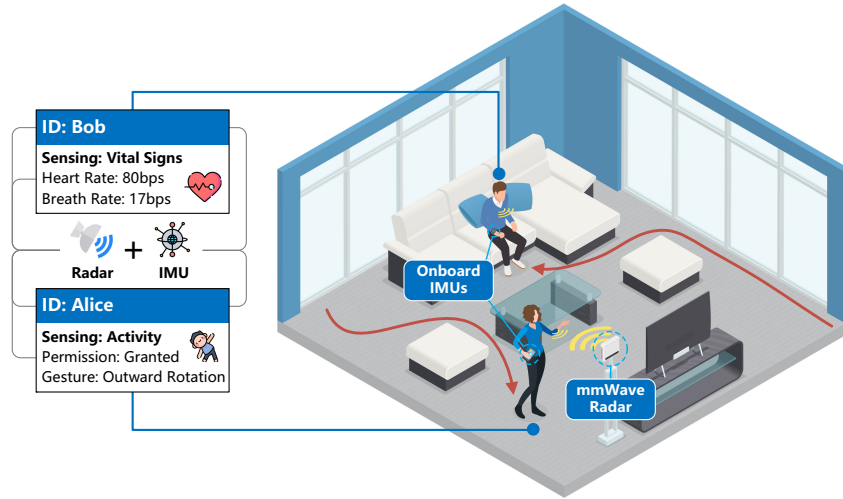


Fig. 1. Personalized tracking endows identity directionality to mmWave sensing systems.

1.4 Challenges & Solutions

We need to address the following four key technical challenges when implementing the PmTrack system.

Weak reflection: in radar signal processing, the static clutter removal conducted to suppress ambient reflections may also remove the reflection signal from stationary humans. Moreover, when the line-of-sight (LoS) path is obstructed, the radar signal is scattered and absorbed by the obstruction object, so the reflection from the obstructed object is greatly attenuated. The weakening of the signal reflection makes the target difficult to detect and may result in trajectory loss.

Point cloud overlap: the ranging and angular resolutions of radar are restricted by its bandwidth and antenna array. When two people are close to each other, their positional differences may exceed the target segmentation capability of the radar, resulting in overlapping spatial point clouds. Incorrect clustering and segmentation of point clouds can affect the trajectory-cluster association, a crucial step of multi-person tracking, and can thereby harm the continuity of trajectories and cause incorrect tracking updates.

Body-bounce ghost: multipath reflections may cause false targets known as "ghosts" in radar detection. Although various ghost suppression methods have been proposed [9, 16, 18, 19, 45], they are difficult to distinguish body-bounce ghosts (*i.e.*, the first-order multipath type-1 defined in [34]) in single-device mmWave tracking because they behave very similarly to the obstructed real targets. Ghosts can not only be detected as independent targets and generate false trajectories, but also interfere with the tracking of real trajectories.

The above challenges, acting independently or together, create a series of practical tracking problems. In this paper, we conduct in-depth research on these problems and propose a comprehensive set of optimization methods, covering various aspects of the tracking process such as detection enhancement, interference suppression, continuity maintenance, and trajectory correction, to guarantee the trajectory accuracy in practical scenarios.

Gimbal lock: when the attitude of the IMU makes the directions of two rotation axes very close, they become linearly correlated and thus reducing the freedom degrees of rotation, which is the so-called gimbal lock. Gimbal lock causes the calculation result of affected Euler angles to fluctuate dramatically over a range of 180° , making it difficult to accurately describe the real attitude state. To address this issue, we analyze the motion representation of Euler angles and propose an angular correction method to counteract the anomalous orientation fluctuation.

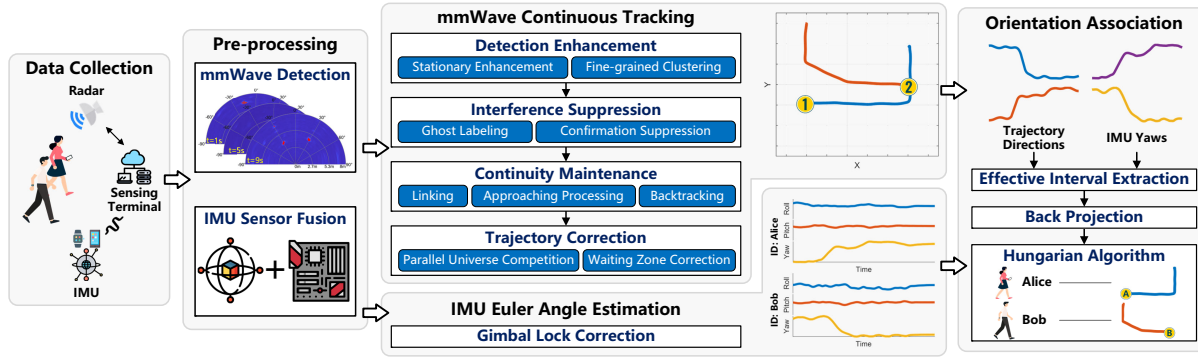


Fig. 2. Overview of the PmTrack system.

1.5 Novelty & Advantages

The novelty of our work is demonstrated in the following three aspects: (1) We propose to use orientation as a matching feature, which overcomes the data heterogeneity between radar and IMU, gets rid of the cumulative error of the accelerometer, and eventually achieves high-accuracy matching. (2) We overcome a series of trajectory errors caused by weak reflection, point cloud overlap, and body-bounce ghost in mmWave radar tracking, ensuring the continuity and accuracy of the trajectory. (3) An orientation correction method for the gimbal lock error is proposed, enabling PmTrack to work reliably with different carrying attitudes of IMU.

The PmTrack system has two main advantages over previous works: (1) Compared with using individual differences, PmTrack avoids the burden of collecting training data and is not affected by user scale, data timeliness, and application environment, making it easy to deploy in various scenarios. (2) Compared with other RF beacons [26, 46, 49, 58, 81], IMU is commonly available in portable devices and transmits data through existing wireless networks. It does not require the installation of additional RF devices and does not impose an additional carrying burden on users. (3) Compared with ImmTrack [11] which adopts the trajectory to associate IMU and radar targets, PmTrack utilizes the orientation that is highly coincident in both systems, resulting in superior matching performance. Experimental results reveal that PmTrack achieves an identification accuracy of 98% and 95% with five people in the hall and meeting room, respectively.

The rest of this paper is structured as follows. In Section 2, we present an overview of the proposed PmTrack system. Section 3 introduces the data collection and pre-processing of mmWave radar and IMU. We describe the basic design of PmTrack in Section 4. Section 5 focuses on solving practical problems in multi-person tracking and identification, and in Section 6 we address the IMU gimbal lock error. Section 7 presents the implementation and evaluation results. We discuss the related works in Section 8, and conclude this paper in Section 9.

2 SYSTEM OVERVIEW

The overview of the PmTrack system is shown in Fig. 2. In the data collection stage, the mmWave radar transceives the FMCW signal and transmits the sampled data to the sensing terminal. The IMUs embedded in portable devices such as smartwatches and smartphones measure accelerations and angular velocities, which are wirelessly transmitted to the sensing terminal by the portable devices. Then, the sensing terminal uses radar data to track the movement trajectories of multiple persons, while utilizing the IMU data for attitude tracking and Euler angle calculation. During the tracking process, we proposed various optimization methods for detection enhancement, interference suppression, continuity maintenance, and trajectory correction, which successfully overcome a series of tracking problems caused by weak reflection, point cloud overlap, and body-bounce ghost. Meanwhile, we

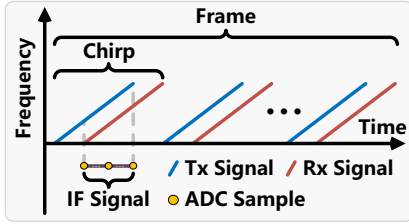


Fig. 3. FMCW mechanism.

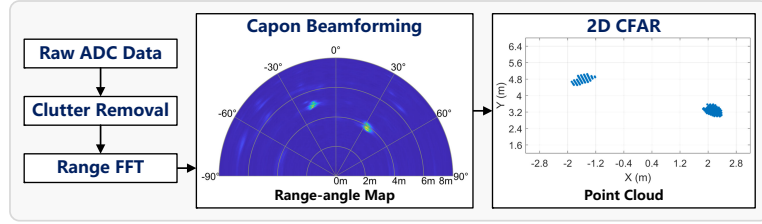


Fig. 4. Range-angle point cloud generation.

propose an orientation correction method to address the gimbal lock error of IMU. The direction of the trajectory and the yaw of the Euler angles represent the orientation of the mmWave radar system and the IMU system, respectively, based on which the back projection method will be applied to calculate the similarity between the two systems. Finally, the Hungarian algorithm is performed to match the trajectories with the IMUs, enabling the radar sensing system to obtain the identity of each target.

3 DATA COLLECTION AND PRE-PROCESSING

This section introduces the working principle and data pre-processing of mmWave radar and IMU, including the FMCW mechanism and point cloud generation of mmWave radar, and the Euler angle calculation of IMU.

3.1 Coordinate System Definition

To avoid potential confusion, we first define the coordinate systems involved in this paper as follows.

Radar/ground coordinate systems: the radar horizontal plane is parallel to the ground in the description and experiment of this paper, so we use radar to establish the ground coordinate system for convenience. Taking radar as the origin, the x-, y-, and z-axes are perpendicular to the radar vertical plane, antenna array plane, and ground, respectively. The trajectory orientation follows the right-hand screw rule along the z-axis.

IMU coordinate system: the local coordinate system of the device. The rotation angles along x-, y-, and z-axes are roll, pitch, and yaw, respectively, and the rule is also the right-hand screw. The rotation order is z-y-x. Specifically, the IMU-defined initial direction of the z-axis is perpendicular to the ground. Therefore, yaw also describes the rotation of IMU along the z-axis in the ground coordinate system.

3.2 mmWave Radar Point Cloud Generation

3.2.1 FMCW Mechanism. The object detection of mmWave radar relies on FMCW chirp signals, whose frequency increases linearly with time, as depicted in Fig. 3. Let the transmitted (Tx) chirp signal be

$$s_t(t) = A_t \exp(j2\pi(f_c + kt)t), \quad (1)$$

where A_t , f_c , and k are the amplitude, start frequency, and slope, respectively. Suppose there is a reflector at distance $D(t) = R + d(t)$ from the radar, where R represents the distance between this Range bin and the radar, and $d(t)$ is the reflector's displacement at time t . Then we have the Rx signal

$$s_r(t) = \alpha s_t(t - 2D(t)/c), \quad (2)$$

where α is the attenuation factor and c is the light speed. Passing the Rx and Tx signals into the mixer, we get the intermediate frequency (IF) signal

$$s(t) = s_t^*(t) s_r(t) \approx A \exp(j4\pi(f_c + kt)D(t)/c) = A \exp(j(2\pi f(t)t + \phi(t))), \quad (3)$$

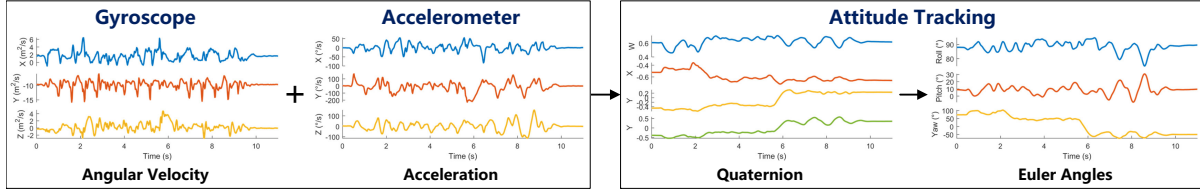


Fig. 5. Simplified sketch of IMU Euler angle calculation.

where A , $f(t) = 2kD(t)/c$, and $\phi(t) = 4\pi f_c D(t)/c$ represent the amplitude, frequency, and phase offset, respectively. Multiple chirps form a frame, which is the basic unit of radar signal processing.

3.2.2 Static Clutter Removal. All objects in the environment reflect radar signals, and the IF signal itself contains strong direct current components. These static components can cause serious interference in detecting moving targets. We use the vector mean-canceling method to remove the static clutters, which calculates the average value of each sampling point or Range bin in a frame and subtracts it from the original signals. This process can greatly improve the signal-to-noise ratio of moving targets while suppressing stationary targets.

3.2.3 Range Detection. By applying fast Fourier transform (FFT), $s(t)$ is transformed from the time domain to the frequency domain, which is known as Range FFT. The frequency components of the IF signal correspond to specific distances, which are referred to as Range bins. The signal of Range bin R can be expressed as

$$s_R(t) = A \exp(j\phi(t)). \quad (4)$$

3.2.4 Angle Estimation. The range-angle (RA) scheme is more suitable for detecting human targets, as it can mitigate the multipath effect and better distinguish targets from the background. We use the Capon beamforming, which accurately estimates the spatial spectrum by minimizing the variance of the output power, thereby effectively suppressing interference signals and improving the target detection capability. Let the channel number of the antenna array be N_a and the chirp number in a frame be N_c . Then, for Range bin R , the antenna array signal S_R is a $N_a \times N_c$ matrix. Next, we can calculate the signal strength from a specific angle θ through

$$P_{R,\theta} = 1/(\mathbf{a}_\theta^H \mathbf{C}_R^{-1} \mathbf{a}_\theta), \quad (5)$$

where $\mathbf{C}_R = S_R S_R^H$ is the covariance matrix that contains the variance information of the signal and noise. $\mathbf{a}_\theta$ is the steering vector, representing the phase difference of the signal from θ arriving at each antenna unit. Using Eq. 5 to calculate the signal strength of each RA bin in the area of interest, we obtain the range-angle map (RAM).

3.2.5 Target Detection. The cell averaging-constant false alarm rate (CA-CFAR) algorithm is then conducted to generate point clouds from RAM, as shown in Fig. 4. The algorithm generates a 2D window around each RA bin, calculates a threshold based on other bins within the window, and determines whether the bin has a target. CA-CFAR adaptively adjusts the threshold and thus is very suitable for dealing with different noise levels.

3.3 IMU Euler Angle Calculation

A 6-axis IMU includes a 3-axis accelerometer and a 3-axis gyroscope, which measures the accelerations and angular velocities, respectively, as shown in Fig. 5. Euler angles describe the attitude relationship between the IMU coordinate system and the ground coordinate system, namely roll, pitch, and yaw, which correspond to rotations around the x-, y-, and z-axes of the IMU, respectively. The same attitude can have different Euler angles in different rotation systems, and this paper adopts the "z-y-x" rotation order. The gyroscope directly measures angular changes, while the accelerometer corrects roll and pitch. In practical deployment, the Kalman filter (KF)

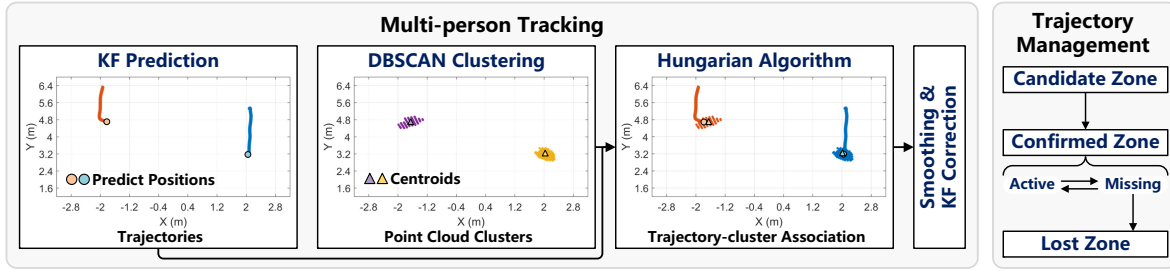


Fig. 6. Multi-person tracking workflow and trajectory management logic.

is often adopted for fusing accelerometer and gyroscope, where the attitude is represented by the quaternion $q = [q_0 \ q_1 \ q_2 \ q_3]$, and the Euler angles can be obtained by

$$\begin{cases} \text{roll} = \text{atan2}(2(q_0q_1 - q_2q_3), 1 - (q_1^2 + q_2^2)) \\ \text{pitch} = \text{asin}(2(q_0q_2 + q_1q_3)) \\ \text{yaw} = \text{atan2}(2(q_0q_3 - q_1q_2), 1 - (q_2^2 + q_3^2)) \end{cases} \quad (6)$$

4 BASIC DESIGN OF PMTRACK

This section presents the basic framework of the PmTrack system. We will first describe the multi-person tracking process with mmWave radar, and then explain how to match the radar-measured trajectories with IMU.

4.1 Multi-person Tracking

The classical point target tracking strategy is introduced, which clusters point clouds and uses the centroids of clusters for trajectory updating, as shown in Fig. 6.

4.1.1 Point Cloud Clustering. Since the person number is usually unknown, and the distribution of human point clouds at different distances and angles differs, the density-based spatial clustering of applications with noise (DBSCAN) algorithm is employed for clustering. DBSCAN groups density-connected points into the same cluster, which is ideal for human point clouds of varying shapes and sizes, and can effectively filter out noisy points. Next, the centroid of each cluster will be calculated with the power as the weight.

4.1.2 Trajectory-cluster Association. The trajectory updates from the association of existing trajectories with new centroids. We calculate the distance of each trajectory-centroid pair and feed it to the Hungarian algorithm as the cost to solve this bipartite graph problem. Based on a distance threshold, the Hungarian algorithm maximizes the number of associations by finding augmented paths and finally outputs the optimal associations.

4.1.3 Trajectory Prediction and Correction. We construct a KF of constant velocity mode for trajectory prediction and correction. At the beginning of each frame's tracking, the KF predicts the trajectory using the state transition matrix from the previous frame to provide a more accurate real-time position for the trajectory-cluster association. KF is also able to update the trajectory in time when it fails to associate for reasons such as occlusion or low reflection strength. After association, the KF corrects the predicted positions with the centroid coordinates.

4.1.4 Trajectory Management. Effective trajectory management is crucial for stable multi-person tracking because clusters may appear in blank positions due to noises and multipath effects, and real targets may fail to be detected due to occlusion and low reflection strength. We adopt the common "candidate-active-missing-lost" logic [8, 56] and modify it as shown in Fig. 6. When a cluster is not associated with any existing trajectory, it may be a new

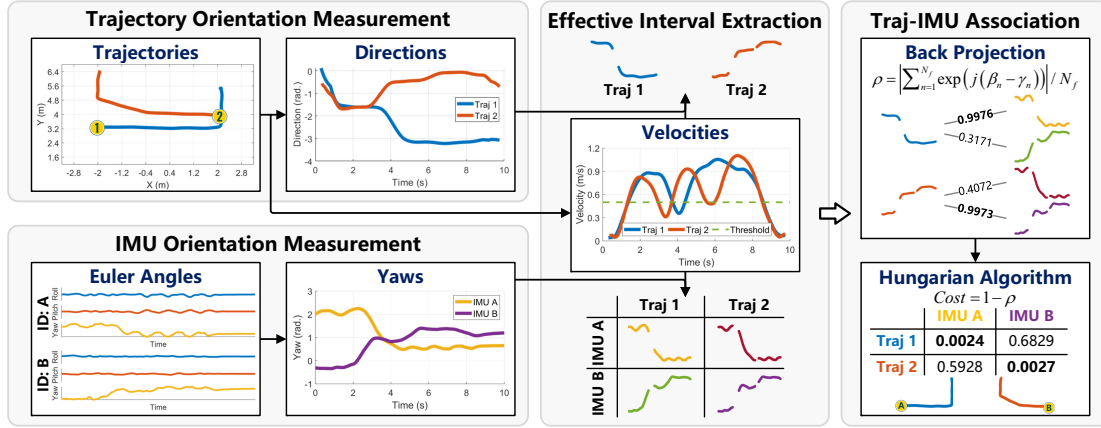


Fig. 7. The trajectory-IMU matching scheme of PmTrack.

target or just noise. So, we create a candidate trajectory and observe if it stably updates. We empirically believe that a target can be considered real if it can be detected in at least 90% of measurements. Specifically, if a candidate trajectory is associated with clusters that are not associated with any existing trajectory in 90% frames within the next 1s, it becomes confirmed. The confirmed zone is the main zone for tracking and it contains active and missing states. An active trajectory will be flagged as missing if it fails to associate with a cluster, and a missing trajectory will be flagged back to active when it re-associates with a cluster. A missing trajectory that has been preserved for 5s will be placed in the lost zone, indicating that it will no longer be involved in the tracking.

4.2 Orientation-based Matching

After radar tracking, PmTrack matches the orientations of the radar and IMU systems using the scheme depicted in Fig. 7 to ultimately identify radar targets.

4.2.1 Orientation Measurement. PmTrack obtains the trajectories' orientations by calculating displacement directions using a moving time window of 0.5s. For IMU, we use the yaw as the orientation. The two orientations will be unwrapped and pass through a 1s-width moving average filter. The IMU orientation is aligned in time with the trajectory orientation through linear interpolation.

4.2.2 Effective Interval Extraction. When a person becomes stationary, the calculation of its trajectory orientation is inaccurate and meaningless, so we use a threshold of 0.5m/s to filter out time intervals where the displacement is too small. Meanwhile, the discrimination between "correct pairs" and "incorrect pairs" will be enhanced by extracting the orientation of each IMU from the effective intervals of each trajectory.

4.2.3 Matching Strategy. Let the number of a trajectory's effective frames be N_f , then we get the trajectory orientation vector $\beta = [\beta_1, \dots, \beta_{N_f}]$ and the IMU orientation vector $\gamma = [\gamma_1, \dots, \gamma_{N_f}]$. PmTrack employs the back projection algorithm for matching, which calculates the similarity between signal vectors and is commonly used in RF imaging and localization [17]. The back projection algorithm can also be used to calculate the similarity between angle vectors, as when we convert an angle β to complex form $\exp(j\beta)$, it is equivalent to a signal with normalized amplitude and phase β . The similarity between the vectors β and γ can be expressed as

$$\rho(\beta, \gamma) = \left| \sum_{n=1}^{N_f} \exp(j(\beta_n - \gamma_n)) \right| / N_f, \quad (7)$$

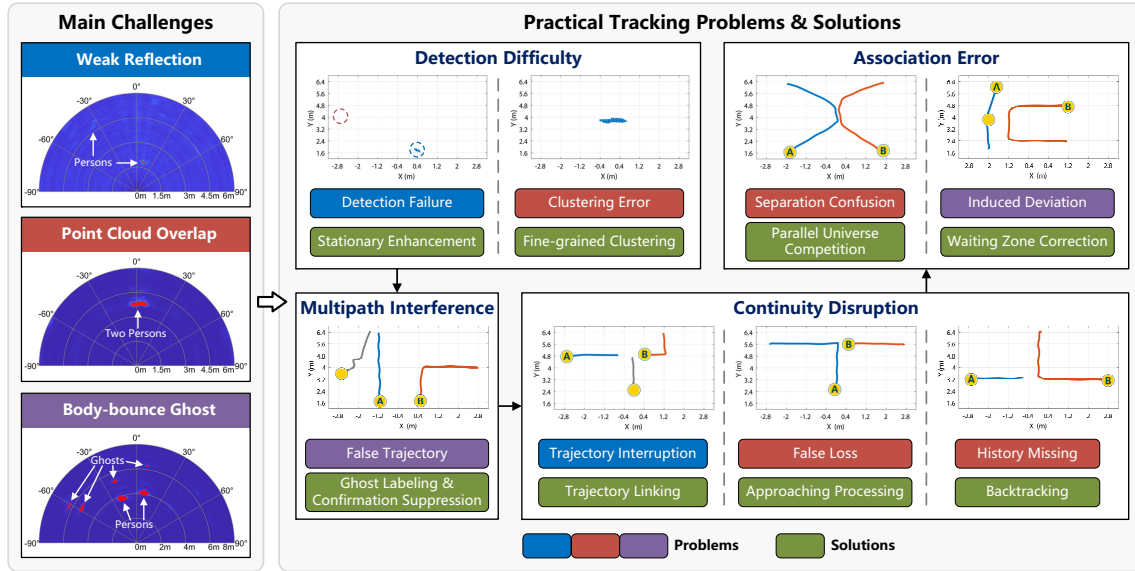


Fig. 8. Practical problems in multi-person tracking and the proposed solutions.

which has advantages in angle vector matching because it naturally overcomes the problems of initial phase difference and cyclic ambiguity [32]. Finally, the similarity between each trajectory and each IMU will be converted to $cost = 1 - \rho$ to serve as the cost input of the Hungarian algorithm for trajectory-IMU association, thus confirming the trajectories' identities. The trajectory-IMU matching does not need to perform at every frame, and in this paper, it performs once every 1s (*i.e.*, 20 frames according to the experimental setup).

5 ADAPTION TO PRACTICAL TRACKING SCENARIOS

The reliable identity association of PmTrack requires radar to provide continuous and accurate walking trajectories. However, radar multi-person tracking faces three major challenges, including weak reflection, point cloud overlap, and body-bounce ghost. As shown in Fig. 8, these three challenges cause various problems in four aspects: detection difficulty, multipath interference, continuity disruption, and association error. In this section, we will introduce these challenges, elaborate on the tracking problems they cause, and our solutions as well.

5.1 Main Challenges in mmWave Human Tracking

5.1.1 Weak Reflection. Radar performs static clutter removal to make the human reflection signal more prominent compared to the ambient reflection signal. However, this removal may also remove the reflection signal from stationary humans, making them difficult to detect, as shown in Fig. 9c. Additionally, when the LoS path is obstructed, the obstruction scatters and absorbs the signal energy, leading to in a decrease in the strength of the target reflection signal. Continuous detection failure not only poses a risk of trajectory loss, but also may trigger various tracking errors along with other problems.

5.1.2 Point Cloud Overlap. The resolution and detection capability of a radar are limited by its frequency, bandwidth, and antenna array aperture. When two people are close to each other, the radar cannot distinguish them as two separate individuals. On the one hand, an inaccurate estimation of the number of individuals can

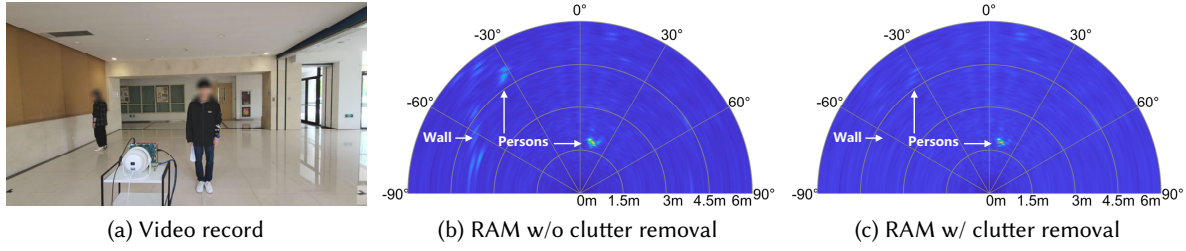


Fig. 9. The static clutter removal also removes the reflection signal from stationary humans.

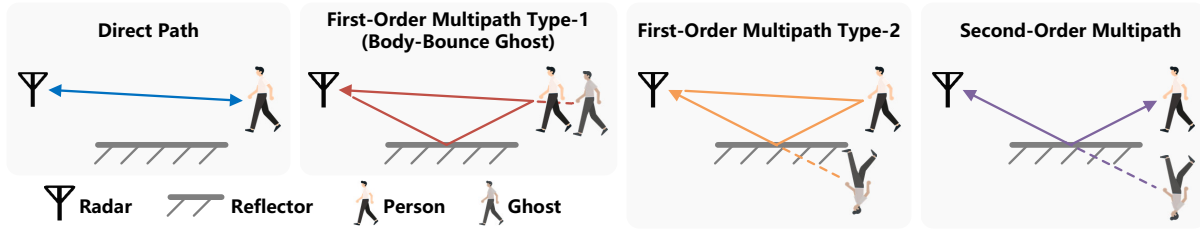


Fig. 10. Different multipath types faced by radar detection.

interfere with the continuity and integrity of trajectories. On the other hand, point cloud overlap can have adverse effects on the association between trajectories and clusters.

5.1.3 Body-bounce Ghost. Due to the presence of strong reflection surfaces such as the floor and walls, the radar signal will propagate from Tx to Rx through several different paths, and the signals of these different paths will be superimposed with the LoS signal at the Rx, which is known as the multipath effect. As illustrated in Fig. 10, the Rx signal includes the following components [16, 34]. (1) Direct path: the straight path between radar and target, on which the radar detection depends. (2) First-order multipath type-1: signal reaches the target via the reflector and then returns to radar through the direct path. (3) First-order multipath type-2: signal reaches the target through the direct path and then returns to radar via the reflector. (4) Second-order multipath: signal reaches the target via the reflector and returns the same way. (5) Higher-order multipath: signal undergoes multiple multipath reflections during propagation. Due to the path loss and reflection loss, the higher-order multipath signals are usually very weak and thus can be ignored. Multipath signals can produce false targets called ghosts in radar detection, as demonstrated in Fig. 11. The ghost of first-order multipath type-2 and second-order multipath usually appear "outside" the reflector, such as outside the walls or below the floor, so they can be easily eliminated by restricting the position of the point cloud. The ghost of first-order multipath type-1, known as "body-bounce ghost", is particularly challenging to deal with because it has all the characteristics of an obscured real person: weak signal strength, reasonable spatial position, contains human body information such as height and shape, and includes motion information.

5.2 Detection Enhancement

We need to improve target detection in two aspects to enhance the stability and accuracy of the tracking system. First, the probability of the target being detected needs to be increased to ensure that the target is not lost. Second, point clouds need to be accurately clustered, especially when they represent adjacent multiple individuals.

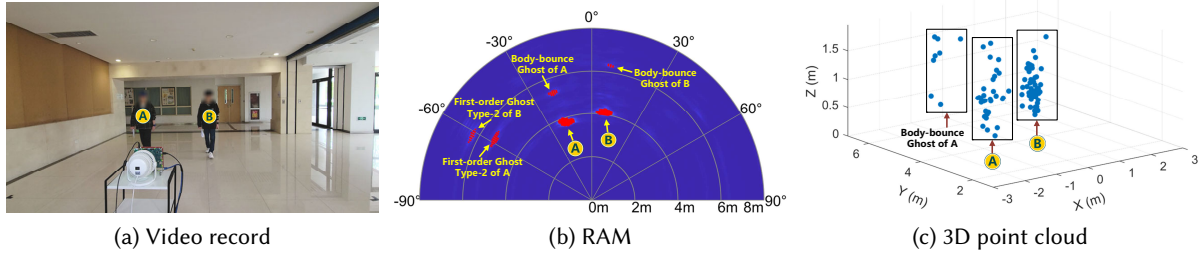


Fig. 11. Examples of ghost targets in radar detection.

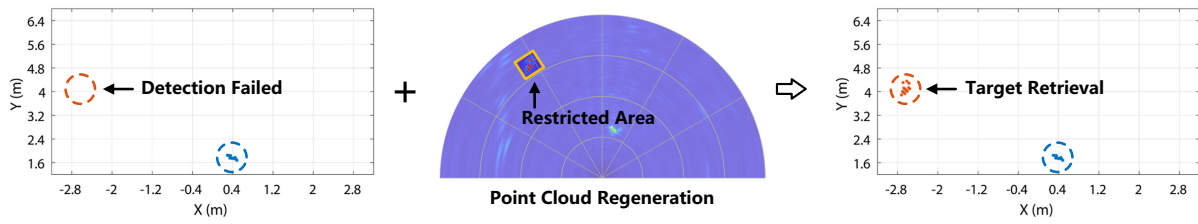


Fig. 12. Sketch of stationary target enhancement.

5.2.1 Stationary Target Enhancement. The weak reflection signal of stationary targets makes them prone to become missing. To reduce the effect of detection failures, we propose a stationary target enhancement method to help retrieve missing targets. If a confirmed trajectory is missing, we will regenerate the point cloud within a surrounding area centered at its last known position, as shown in Fig. 12. In the point cloud regeneration, static clutter removal will not be performed and CFAR parameters will be more relaxed. The new point cloud will be added to the existing point cloud for re-clustering. It should be noted that the point cloud generation, particularly Capon beamforming and CFAR calculations, is a time-consuming process. So on the one hand, it is necessary to restrict the detection area to the surrounding area of the target, and on the other hand, the using frequency of stationary target enhancement should be reduced. In this paper, the operation is set with a calm window of 1s, which balances a low computational burden with the timeliness of target retrieval.

5.2.2 Fine-grained Clustering. When the point clouds of two individuals overlap, DBSCAN will classify their point clouds as belonging to a single individual, as shown in Fig. 13b. Therefore, we conduct fine-grained clustering, which determines whether a cluster contains multiple people based on the cluster diameter. We draw inspiration from the idea of using Gaussian mixture model (GMM) clustering to refine the results of DBSCAN [4], and incorporate this approach into our system with a more detailed design. The most intuitive difference between GMM and DBSCAN is that the number of clusters in GMM is determined as an input at the beginning. GMM assumes that clusters are composed of multiple Gaussian distributions. It calculates each point a probability value for belonging to each Gaussian distribution and assigns the point to the cluster with the highest probability. The contour of the overlapping point clouds can often be viewed as two intersecting ellipses. In such cases, GMM can effectively separate the two ellipses by modeling them as two separate Gaussian distributions. In PmTrack system, we obtain the convex hull of the point cloud using the Quickhull algorithm, calculate distances between the vertices, and use the maximum value as the diameter of the point cloud. When the diameter exceeds 0.9m, the cluster is considered to contain multiple people, and GMM clustering will be performed to segment the point

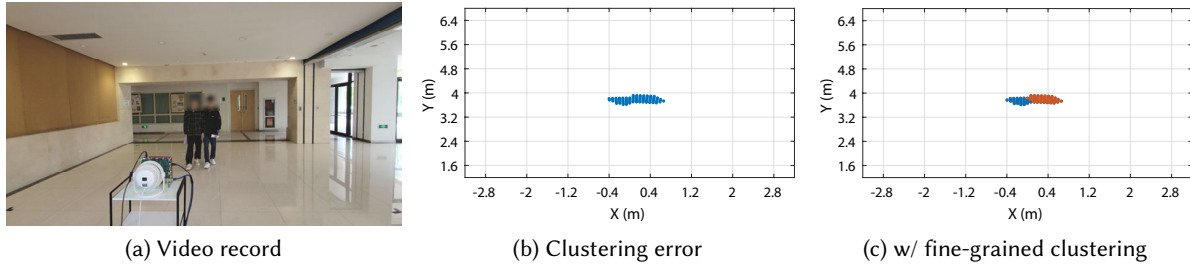


Fig. 13. Example of clustering error and fine-grained clustering.

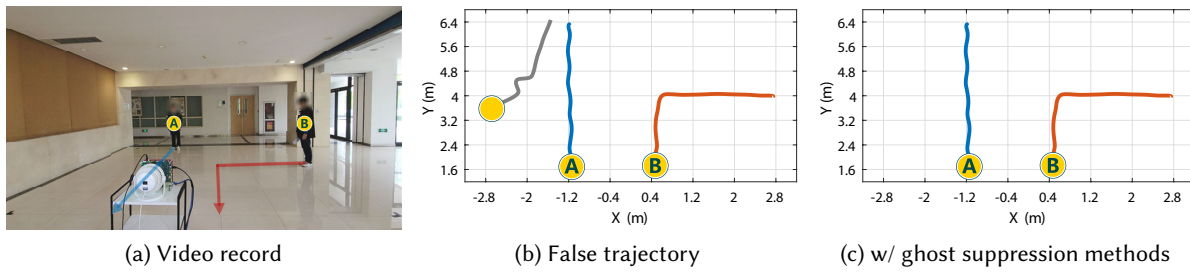


Fig. 14. Example of false trajectory and ghost suppression methods.

cloud, in which the number of clusters is calculated by dividing the diameter of the point cloud by 0.7m and rounding up to the nearest integer, and the initial positions are the equally spaced points on the diameter line. In this step, we use a diameter threshold significantly larger than the shoulder width (typically less than 0.4m). This is because the diameter of a point cloud is affected by many factors such as signal strength, CFAR parameters, and radar detection resolution, and may not be close to the shoulder width. As shown in Fig. 13c, the point clouds of the two individuals are successfully separated through fine-grained clustering, which helps the continuity and integrity of tracking results.

5.3 Interference Suppression

Multipath interference affects the tracking system in the form of a point cloud, which can cause tracking errors and even chaos. The most direct impact caused by ghosts is the generation of false trajectories. Although the ghost behaves with some uncertainty, it may continuously present for a while, which causes the tracking system to identify it as a new target, as shown in Fig. 14b. Existing methods cannot accurately identify body bounce ghosts and real targets, and incorrect discrimination can lead to the loss of real trajectories.

5.3.1 Ghost Labeling. To minimize the generation of false trajectories while ensuring that the normal tracking process is not affected, we propose ghost labeling and confirmation suppression. Ghost labeling means that instead of judging whether a cluster is a ghost or not, we label all clusters that may be ghosts to support subsequent processing, which provides flexibility and fault tolerance for the tracking system. The judgment of the likelihood to be a ghost is determined based on the angular consistency of the ghost with its real target, *i.e.*, the ghost is always behind the real target. However, in practical measurements, the centroid and the point clouds of real and ghost clusters may be offset in angles, so we design the logic as shown in Fig. 15. For a cluster C_1 with a

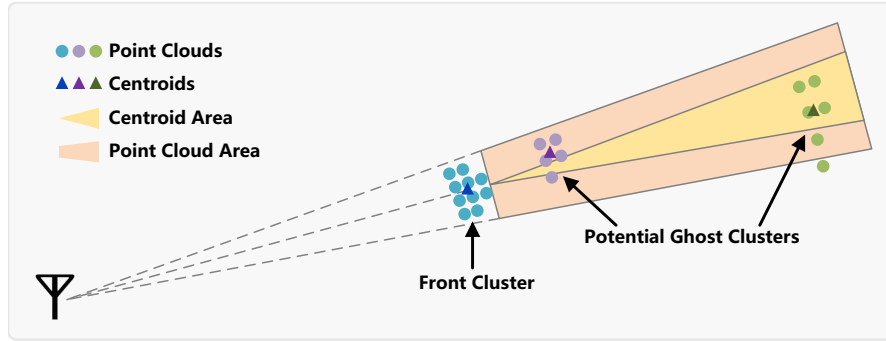


Fig. 15. Sketch of the ghost labeling mechanism.

centroid-radar distance D , we search whether there is a cluster C_2 within the range of $(D, 2D)$ that satisfies either of the following two conditions. (1) The angle between lines C_1 - C_2 and C_1 -radar is less than 5° . The centroid area that satisfies this condition is represented in yellow in Fig. 15. (2) All the points of C_2 are within the angular limit of C_1 's point cloud. The point cloud area that meets this condition is represented in red. Clusters that meet any of the conditions are considered potential ghost clusters and are labeled.

5.3.2 Confirmation Suppression. Once a cluster is labeled as having ghost suspicions, it still participates in the tracking process as normal, but its transition from the candidate zone to the confirmed zone is suppressed: a candidate trajectory can only be confirmed if all associated clusters in the last 0.5s are non-ghost clusters. Through this soft decision-making approach, we are able to prevent the creation of false trajectories without affecting normal association, as shown in Fig. 14c.

5.4 Continuity Maintenance

Although our proposed enhancement methods can significantly improve the success rate and accuracy of detection, they are still insufficient to avoid detection failures. When the target is occluded, point cloud re-generation may be unable to enhance its reflection. In addition, to prevent false overlapping alarms and even false trajectories, the diameter threshold in fine-grained clustering cannot be too low. However, a higher threshold may fail to distinguish overlapping point clouds and segment them. Next, we will explain how PmTrack maintains trajectory continuity when detection failures occur.

5.4.1 Trajectory Linking. The occlusion of the LoS path can have two levels of impact. First, the reflection signal may be too weak to detect the target. Second, even if the point cloud can be generated and clustered, the point cloud may be incomplete cause in errors in the centroid measurement. As a result, when the point cloud of the person reappears, the true position (*i.e.*, the new cluster's centroid) may have a significant difference from the position predicted by the KF, therefore the trajectory-cluster association may fail. This cluster that fails the association will go to the candidate zone and finally generate a new trajectory, while the old trajectory is interrupted. In Fig. 16b, in addition to the trajectory interruption of B, the identification result of A is incorrect due to the presence of the new trajectory. We solve this problem by trajectory linking. Before a candidate trajectory is transferred to the confirmed zone, we extract its starting position and timestamp, and search whether there is an active trajectory that becomes missing near this position and timestamp and has not yet returned to the active state. If there is, the candidate trajectory will be linked to the missing trajectory. As shown in Fig. 16c, when the new trajectory is generated, it observes that trajectory B enters the missing state near its position and time origin. Therefore, this new trajectory is linked to trajectory B, thus avoiding its interruption.

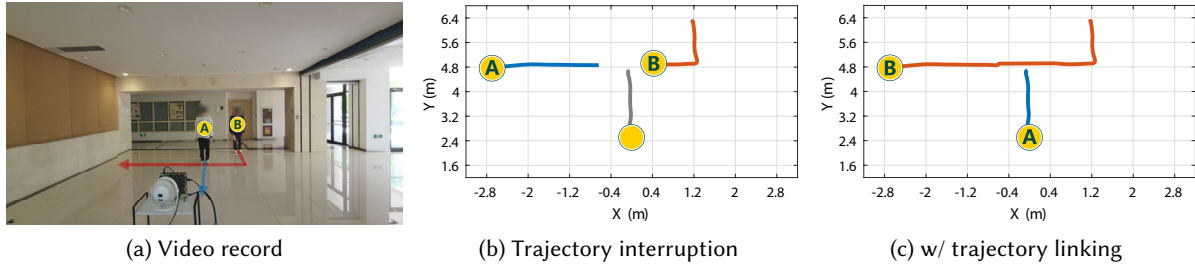


Fig. 16. Example of trajectory interruption and linking.

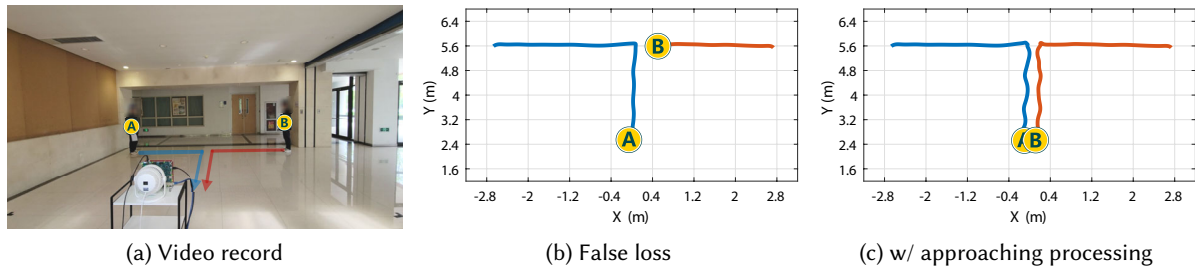


Fig. 17. Example of false loss and approaching processing.

5.4.2 Approaching Processing. A common scenario is that two individuals approach and walk together. When two point clouds overlap, they are clustered into a single cluster. Therefore, if the fine-grained clustering fails in the subsequent tracking, only one trajectory is updated while the other becomes missing and will finally become lost, as shown in Fig. 17b. PmTrack implements an optimization method called approaching processing to robustly solve this problem. When a trajectory $\mathcal{T}$ fails the association, the system will search whether there is a cluster associated with other trajectories that meets one of the following conditions: (1) The $\mathcal{T}$ -centroid distance is less than 0.6m. (2) The minimum distance between $\mathcal{T}$ and the cluster's points is less than 0.1m. (3) The $\mathcal{T}$ -centroid distance is less than 0.8m, and the minimum distance between $\mathcal{T}$ and the cluster's points is less than 0.1m. If any of these conditions are met, $\mathcal{T}$ and the cluster will be added to a temporary overlap record. Finally, taking the number of overlapping trajectories as the number of clusters and the final positions of the trajectories as the initial positions, GMM clustering is performed on each overlapping cluster to create sufficient clusters and ensure correct association, as shown in Fig. 17c. Approaching processing triggers GMM re-clustering in a mandatory manner with more accurate initial parameters, thus effectively preventing the false loss issue when fine-grained clustering goes wrong.

5.4.3 Trajectory Backtracking. Another case is that two people walk into the radar field of view (FoV) together and their point clouds overlap at the beginning. Due to the lack of detection on the trajectory approaching, the point cloud will be treated as coming from only one person. When the two persons separate, one of them will be tracked as a new trajectory that loses its history, as shown in Fig. 18b. The fine-grained clustering cannot guarantee the segmentation of overlapping targets, so we ensure the retrieval of the history trajectory by another step, called trajectory backtracking. Before a candidate trajectory is transferred to the confirmed zone, it loads past clustering results and performs backward tracking along the time axis until there are consecutive association

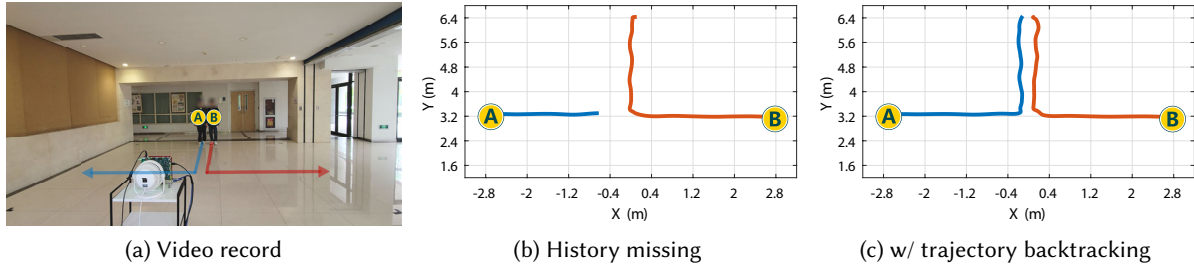


Fig. 18. Example of history missing and trajectory backtracking.

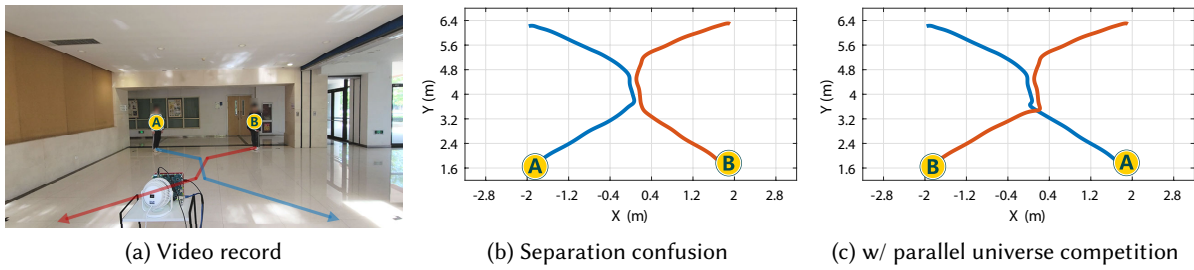


Fig. 19. Example of separation confusion and parallel universe competition.

failures of 0.5s. During the backtracking, PmTrack can perform approaching processing along the reverse time axis, as shown in Fig. 18c. The fine-grained clustering can suppress errors in the number of targets in the stage of walking together, while the backtracking further ensures the integrity of the trajectory.

5.5 Trajectory Correction

The tracking process relies on the association between trajectories and clusters, but association errors can occur. Next, we will introduce how PmTrack corrects trajectory errors that arise from incorrect associations.

5.5.1 Parallel Universe Competition. Although the above methods ensure the trajectory continuity of adjacent people, the association between trajectories and clusters may be confused when the adjacent people separate. GMM clustering can segment a cluster into multiple clusters, but it cannot guarantee segmentation accuracy. When overlapping point clouds cannot be clearly presented as multiple 2D Gaussian distributions, the results of GMM will deviate from the ground truth. Even if the clustering results are correct, interaction and positional exchange between two individuals may still cause association errors, as shown in Fig. 19b. Given that this separation confusion is difficult to immediately distinguish, we propose a parallel universe competition mechanism, which aims to establish a parallel universe for each possible trajectory-cluster association and select the retained universe through subsequent analysis. Specifically, the mechanism includes the following processes.

(1) After the tracking of a frame is completed, trajectories within a distance of less than 0.4m from each other are recorded in the overlap record of that frame. (2) Compare the overlap record with the previous frame and observe if any overlap groups have changed. If so, record the involved trajectories in the separation record. (3) Create parallel universes based on separation records. For example, if a separation group consists of trajectories $\mathcal{T}_1$ and $\mathcal{T}_2$, the latest positions of them in the main universe will be assigned to trajectories " $\mathcal{T}_1$ and $\mathcal{T}_2$ " and " $\mathcal{T}_2$

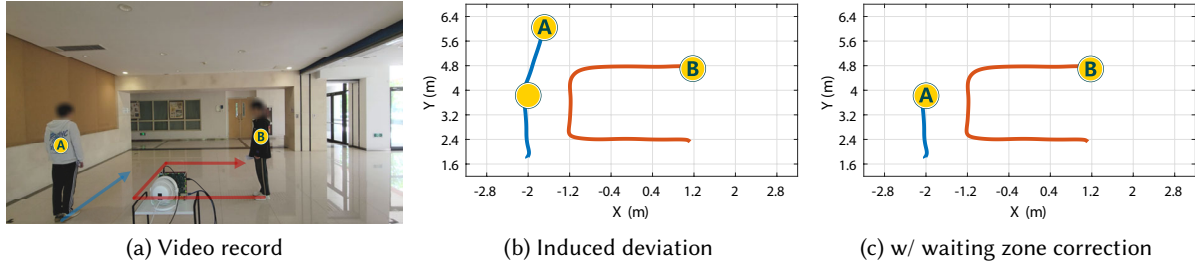


Fig. 20. Example of induced deviation and waiting zone correction.

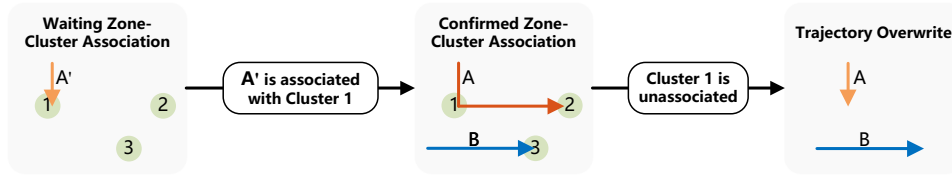


Fig. 21. Waiting zone correction. The deviated trajectory A induced by the ghost of B is corrected to its true position.

and $\mathcal{T}_1''$ respectively in the newly created two parallel universes. When separation occurs again, each universe will generate new parallel universes. The child universes inherit the relationship between the parent universe and the main universe and makes changes based on it. In subsequent tracking, the latest position of trajectories in the main universe will be assigned to the corresponding trajectories in each parallel universe. (4) Input all universes to the orientation matching with IMU, and calculate the average cost of each universe according to its trajectory-cluster association result. (5) Finally, select the universe with the lowest sum cost as the main universe and delete universes with very high cost. Fig. 19c shows an example of trajectory association correction through the parallel universe competition mechanism.

5.5.2 Waiting Zone Correction. The induced deviation is caused by multiple factors. The point cloud of a stationary or obscured person may be insufficient, therefore their trajectory may be induced if there is another person's ghost cluster around him. For example, in Fig. 20b the trajectory of A is induced by the ghost of B, while at the real position of A, an independent target generates afterwards. The difficulty of avoiding such ambiguity is that it is not clear whether trajectory A is deviated or it is on the correct path. We propose a waiting zone correction mechanism to flexibly deal with this problem. When a trajectory associates with a cluster with ghost suspicions, its last position will be recorded in the waiting zone. In each frame of the next period of time, waiting trajectories will determine whether they should overwrite the corresponding confirmed trajectories. We use the example shown in Fig. 21 to introduce the correction process, where trajectory A is induced by the ghost of B. The position of A when it becomes induced is recorded as the waiting trajectory A'. In a frame after A becomes induced, three point cloud clusters are detected, and clusters 1, 2, and 3 represent A's true position, A's induced position (B's ghost), and B's position. Through ghost labeling, cluster 2 has ghost suspicion, while clusters 1 and 3 do not. (1) Independent of the confirmed zone, the waiting zone is associated with the clusters, resulting in A' being associated with cluster 1. At this point, cluster 1 may be the true position of A or it may belong to someone else. (2) Then, PmTrack checks whether cluster 1 is associated with any confirmed trajectory, and the result is negative. It means that cluster 1 can only represent the reappearance of A. (3) Use A' to update

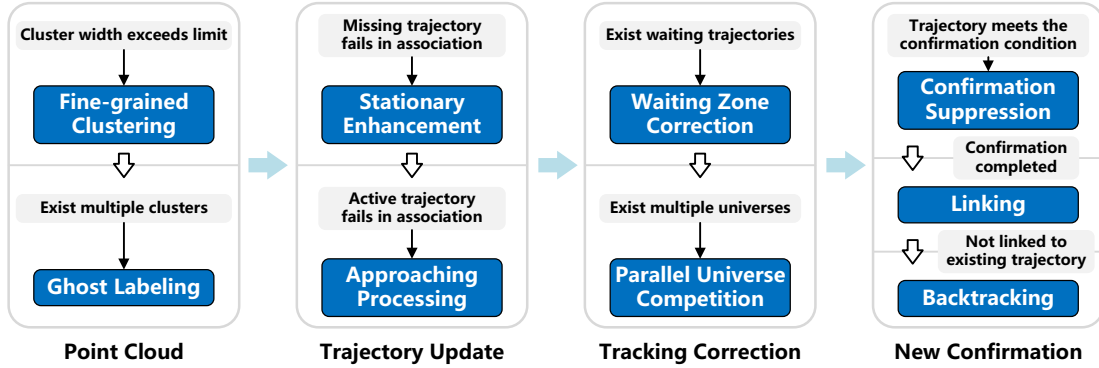


Fig. 22. Summary of optimization methods in multi-person tracking.

the trajectory of A and delete its induced tracking history. Each waiting trajectory will exist for 3s or until its corresponding confirmed trajectory no longer has ghost suspicions. The waiting zone correction mechanism can effectively pull the deviated trajectories back to their correct positions, as shown in Fig. 20c, and the stationary target enhancement further improves the probability of the targets being redetected.

5.6 Summary

PmTrack enables all optimization methods by default, and the execution of each module depends on the trigger conditions. We summarize the triggers and the order in Fig. 22. In general, PmTrack sequentially determines whether to execute the corresponding method during the tracking process of each frame. The optimization methods usually work independently, and sometimes they collaborate with other methods. We have carefully designed and optimized the combination of these solutions to ensure their time efficiency and effectiveness. It should be noted that trajectory linking, backtracking, and overwriting will change the tracking history. At this time, the overlap record, separation record, and parallel universe record will all rewind to the corresponding frame. Then, these records will be re-recorded and updated frame by frame until the current frame. In addition, the relationship between adjacent trajectories may repeatedly switch between overlap and separation, so we set a cooldown window for separation records. For a separation group, if the same group exists in the previous 2s, the previous record will be deleted, and the parallel universes will backtrack and be re-generated.

6 OVERCOMING IMU GIMBAL LOCK

Due to the data heterogeneity, the information from the radar system is not suitable for directly matching with the quaternion or rotation matrix of the IMU system. Therefore, PmTrack calculates IMU yaw from the quaternion and matches it with the trajectory direction. However, calculating yaw from quaternions may introduce errors. In 3D space, the rotation of an object can be expressed as a 3×3 rotation matrix, where each column corresponds to a rotation axis. The rotation matrix must satisfy orthogonality and a determinant of 1, meaning that each column must be a unit vector and mutually orthogonal. When the directions of two rotation axes are very close, their directions become linearly dependent, causing the rotation matrix to lose a column or a row, which reduces the degree of freedom of rotation. This is the gimbal lock problem. Mathematically, the gimbal lock can cause the rotation matrix to become singular, leading to the calculation error of Euler angles. Specifically, when the pitch of the IMU is close to $\pm 90^\circ$, the directions of the yaw axis and the roll axis become very close, resulting in the two columns of the rotation matrix being linearly dependent, *i.e.*, they are collinear or in the reverse direction. This is a common phenomenon, for example, the natural drooping of a person's arm may cause the IMU in

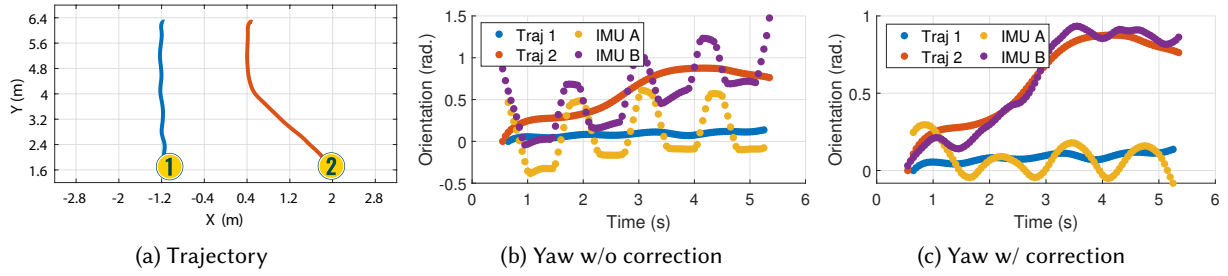


Fig. 23. Example of gimbal lock error and orientation correction.

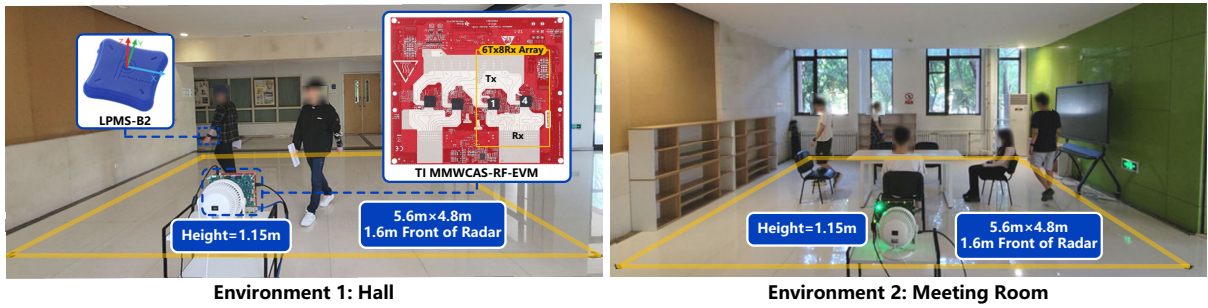


Fig. 24. Experimental environments.

a smartwatch to experience the gimbal lock. Fig. 23b shows the effect of the gimbal lock on IMU orientation measurement, where the continuous and drastic changes in yaw make it difficult to accurately describe the direction of a person's movement. To overcome this problem, we review the physical meaning of each Euler angle. When gimbal lock occurs, the directions of roll and yaw coincide, causing these two angles to change in the same trend. On the other hand, we notice that the actual roll hardly changes during this time because it represents the angle at which a person raises the arm to the side, which is an unusual action for a person who is walking. Therefore, we use roll as a benchmark to correct yaw by

$$\text{yaw}' = \begin{cases} \text{yaw} - (\text{roll} - \text{roll}_1), & \text{percentile}(\text{pitch}, 25) > 45^\circ \\ \text{yaw} + (\text{roll} - \text{roll}_1), & \text{percentile}(\text{pitch}, 75) < -45^\circ \end{cases} \quad (8)$$

where **yaw**, **roll**, and **pitch** represent the Euler angle vectors, and roll_1 is the first element of **roll**. Fig. 23c shows the IMU orientation after correction. Although there are still fluctuations, they can already correspond well to the actual moving orientations.

7 PERFORMANCE EVALUATION

In this section, we conduct extensive experiments in two indoor scenarios to comprehensively evaluate the performance of PmTrack. Each participant is fully informed about the experimental content and purpose.

7.1 Setup

7.1.1 Environment. The experimental environments are shown in Fig. 24. The first environment is a hall with a wall on one side, and the other environment simulates a meeting room. In both environments, we use tiles with a

Table 1. mmWave Radar Parameters

Parameter	Start Freq	Slope	Sample Rate	Sample Num	Chirp Period	Chirp Num	Frame Period
Value	77GHz	99MHz/us	7.04MHz	256	50us ($\times 6Tx$)	128 ($\times 6Tx$)	50ms

side length of 0.8m as a reference for area division and trajectory setting. The walking area is $5.6m \times 4.8m$, the distance between the radar and the walking area is 1.6m, and the height of the radar is 1.15m.

7.1.2 Hardware. mmWave radar: the TI MMWCAS-RF-EVM [55] is deployed for mmWave radar signal collection. The enabled antenna array is 6Tx8Rx, and the specific parameters are listed in Table 1. Since PmTrack only requires planar point clouds, only data from chip 4's Tx is used, while data from chip 1's Tx is considered as potential supplements. That is, the system actually uses an 11-wavelength virtual linear array formed by 3Tx8Rx.

IMU: we adopt LPMS-B2 [37], a wireless 9-axis IMU. The device includes a magnetometer, which is disabled during the experiments so that it acts as a normal 6-axis IMU. The attitude tracking is carried out on the IMU onboard processor, and the data is transmitted to the laptop via Bluetooth at 50Hz. The IMUs are attached to wristbands and worn on the participants' wrists, and the participants are asked to walk in their usual way and swing their arms in a normal manner.

7.1.3 Software. The data collection of the mmWave radar and IMU is controlled by the official softwares, mmWave Studio and LpmsControl, respectively, and subsequent data processing and algorithm implementation are conducted on MATLAB 2022b.

7.1.4 Metric. Identification accuracy (ID Acc): the purpose of this paper is to identify each detected target of radar, so the most important metric is the identification accuracy, which is the ratio of the number of correctly identified individuals to the total number of individuals.

Trajectory accuracy (Traj Acc): we use trajectory accuracy to evaluate whether a trajectory matches the true trajectory of the target. The metric is the ratio of the number of correctly tracked individuals to the total number of individuals. Please note that if a person's trajectory error occurs at the end of the trajectory, their identification result will be recorded as incorrect. If the end of the trajectory is correct, even if the previous part has errors, the identification may still be correct.

Similarity: the similarity between the direction of trajectory and the yaw of the IMU reflects the rationality and feasibility of PmTrack. The higher the degree of coincidence between the two orientation data, the closer the ρ calculated by Eq. 7 is to 1. Specifically, we take the average of the similarity between each trajectory and its corresponding IMU in the association results of the Hungarian algorithm.

Cost ratio: another important metric is whether there is sufficient discrimination between the correct pairs and the incorrect pairs. In this paper, the similarity ρ is converted to $cost = 1 - \rho$ and then input to the Hungarian algorithm. We use the cost ratio, *i.e.*, the average ratio of the costs of the optimal pairs and the suboptimal pairs, as the metric, where the suboptimal pairs represent the optimal associations after ignoring the original optimal associations. The smaller this value, the greater the distinguishability of matching data.

7.2 Overall Evaluation under Predetermined Trajectories

In this section, we implement predetermined trajectories in the hall for the overall evaluation of PmTrack's basic performance. The trajectory settings are shown in Fig. 25, based on which we set multiple variables. For each setting and variable, we conduct 10 trials, each lasting for 10s.

7.2.1 Influence of Trajectory Difference. We first test the four most basic scenarios: (1) one person walks straight while the other turns; (2) both persons turn, but in different directions; (3) both persons turn in the same

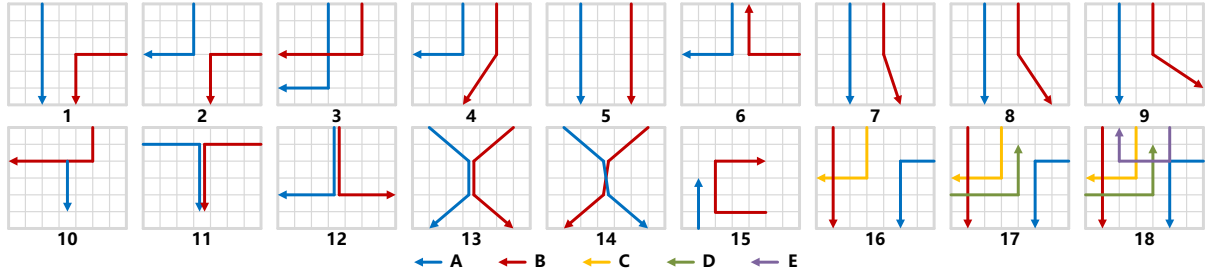


Fig. 25. Predetermined trajectory settings.

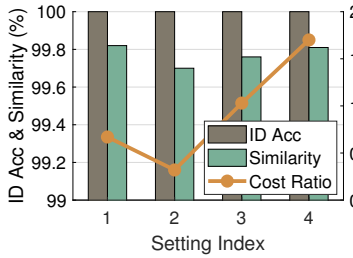


Fig. 26. Basic evaluation.

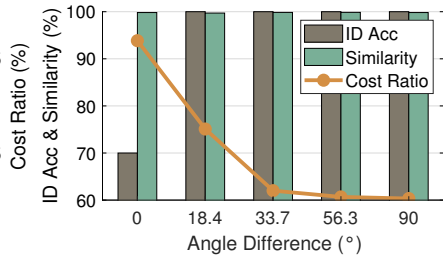


Fig. 27. Influence of trajectory difference.

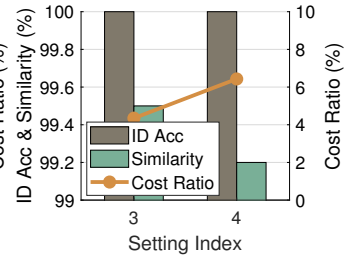


Fig. 28. Strong arm swinging.

direction, but at different times; (4) both persons turn in similar directions, but at different angles. These scenarios correspond to trajectory settings 1~4. The results are shown in Fig. 26. PmTrack achieves 100% identification accuracy thanks to two factors. The orientation similarity between the trajectory and the IMU is high, all greater than 99.7%, while the cost ratio is less than 2%, which determines that the system hardly experiences identity association errors and proves the effectiveness of PmTrack's basic idea and processes.

It appears that PmTrack consistently performs well when there are significant differences between the two trajectories. However, what if the two trajectories are identical? Or, what level of trajectory difference is required for PmTrack to achieve excellent performance? We conduct experiments under settings 5~9, where settings 5 and 6 provide two identical trajectories for straight walking or turning, and settings 7, 8, and 9 provide trajectory differences of varying deviation angles. The results are shown in Fig. 27, where the data of 90° comes from setting 1. When the trajectories are identical, the cost ratio is close to 70%, indicating very low discrimination between correct and incorrect pairs. Therefore, the identification accuracy is only 70%, which is to be expected. When the angle difference reaches 18.4°, the cost ratio drops to 30%, and the identification accuracy reaches 100%, indicating that PmTrack can accurately perform identification as long as small differences exist between trajectories.

7.2.2 Influence of Arm Swinging. During experiments, participants are instructed to swing their arms naturally. Next, we test whether increasing the swinging amplitude would lead to identification errors. Under settings 3 and 4, participants are instructed to swing their arms with a large amplitude, and the experimental results are shown in Fig. 28. Compared to normal swinging, the similarity under strong swinging has decreased but is still greater than 99%, and the cost ratio is below 7%, so it is still easy to achieve 100% identification accuracy. The normal front-and-back arm swinging does not cause the IMU to rotate along the z-axis of the ground coordinate system. Even if the angle changes slightly, this periodic fluctuation will be suppressed by the moving average filtering. Thus, the arm swinging has a negligible impact on the IMU data required for PmTrack.



Fig. 29. Influence of carrying manner.

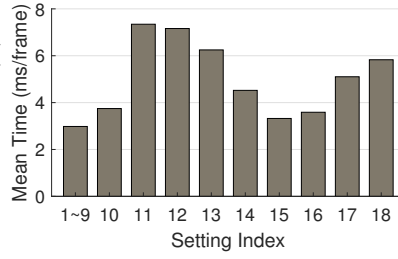


Fig. 30. Time cost per frame.

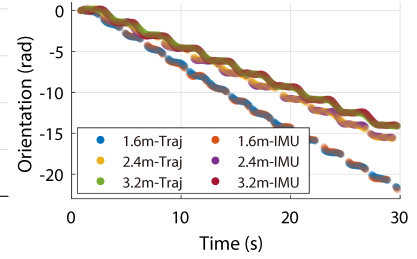


Fig. 31. Orientations over time.

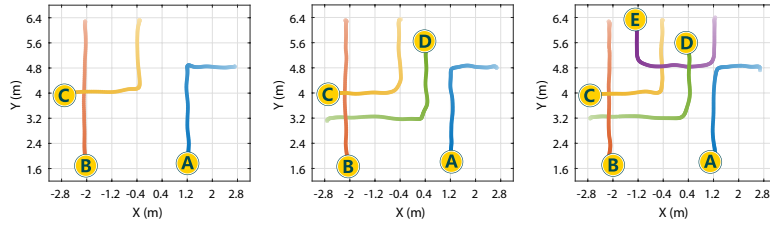


Fig. 32. Part of the results under settings 16~18.

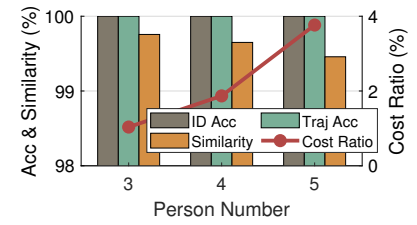


Fig. 33. Metrics under more individuals.

7.2.3 Influence of IMU Carrying Manner. Portable devices, especially smartphones, can be carried in different manners. Under setting 8, participants conduct two additional carrying manners. One manner is to attach the IMU to their phones and walk while looking at the phone, while the other manner is to place the IMU in the pant pocket. It can be seen from Fig. 29 that the similarity is greater than 99.6% in all carrying manners. The cost ratio is less than 4% for the wrist and handheld carrying manners, while this metric is close to 9% for the pant pocket carrying manner. Nevertheless, considering that the difference between the two trajectories is small, these results still demonstrate the effectiveness of PmTrack under various IMU carrying manners.

7.2.4 Performance under More Individuals. We invite more participants to evaluate the performance of PmTrack in more complex scenarios, and the participants walk according to settings 16, 17, and 18, in which various adverse factors such as occlusion and overlap are included. Fig. 32 shows some examples of the tracking and identification results, and detailed metrics are recorded in Fig. 33. The distinguishability of orientation remains significant even in the 5-person scenario.

7.2.5 Time Cost. Next we focus on the time efficiency of PmTrack to evaluate whether it can meet the real-time requirements in practical deployment. In the practical deployment of radar sensing systems, the radar front-end can generate point clouds through specialized onboard processors and output them to the sensing back-end in real-time [56]. Therefore, we only consider the computation time for tracking and identification processes. The CPU of the computer is Intel Core i5-13600KF. All optimization methods are enabled, and the execution of each module depends on the trigger conditions in each frame. Fig. 30 records the mean time cost of one frame under each setting, where the frame period is 50ms. Most 2-person settings take about 3ms. When point cloud overlap occurs, re-clustering significantly increases the time, but it remains below 8ms. Increasing the number of people also affects the computational time, but for the 5-person scenario, the time cost is still less than 6ms. These results represent the overall system delay after receiving the point cloud data from the radar front-end in each frame, demonstrating the time efficiency of PmTrack and its ability to support real-time applications.

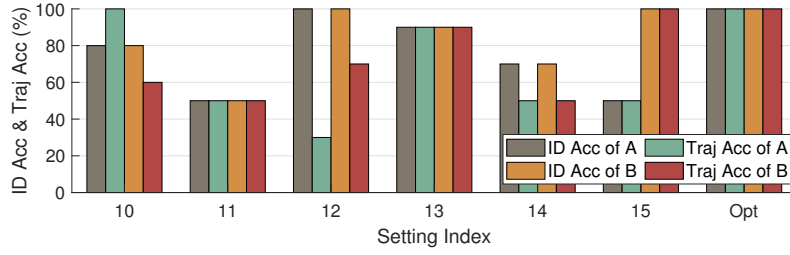


Fig. 34. Metrics after disabling corresponding optimization methods.

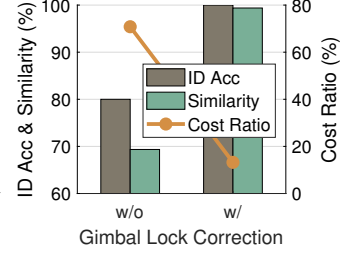


Fig. 35. Gimbal lock correction.

7.2.6 Similarity over Time. Since a 6-axis IMU cannot self-calibrate the cumulative error of yaw, we test whether it affects the similarity between trajectories and IMU. A participant walks continuously along square paths with side lengths of 1.6m, 2.4m, and 3.2m, respectively, during which the orientations of trajectory and IMU are plotted in Fig. 31. It can be seen that the frequent turning within 30s does not cause conspicuous cumulative errors of yaw, and the similarity remained roughly above 99%. Such a long time span is fully sufficient for identification, so we believe that the cumulative error of the IMU would not affect PmTrack, which also indicates the high reliability of our selected matching feature.

7.3 Effectiveness Evaluation of Solutions on Practical Difficulties

PmTrack proposes a series of optimization methods to address various problems caused by weak reflection, point cloud overlap, body-bounce ghost reflection, and gimbal lock. In this section, we will evaluate the effectiveness of these methods. In Sections 7.3.1 to 7.3.5, we disable the optimization methods corresponding to each tracking problem under specific experimental settings. In settings 10, 11, 12, 13&14, and 15, we disable trajectory linking, approaching processing, trajectory backtracking, parallel universe competition, and waiting zone correction, respectively. Three optimization methods are not evaluated. First, the stationary target enhancement is always enabled because it promotes the retrieval of missing targets without side effects. Second, the ghost suppression methods are also enabled to ensure the stability of the tracking process. Third, the fine-grained clustering is disabled in settings 11 and 12 because we prefer to focus on the performance of approaching processing and trajectory backtracking when dealing with overlapping point clouds. Fig. 34 records the average identification accuracy and trajectory accuracy of these experiments. The horizontal label "Opt" refers to the result of PmTrack with all optimization methods enabled in all experiments, which shows 100% accuracy in both metrics.

7.3.1 Trajectory Linking. When a walking person is occluded, their trajectory may be interrupted due to the continuous lack of point clouds. In the ten experiments under setting 10, B's tracking result was interrupted behind A four times, and two of them cause simultaneous identification errors for both A and B. Our proposed trajectory linking method avoids such interruptions and effectively maintains the continuity of the trajectory.

7.3.2 Approaching Processing. When two people come together as in setting 11, their point clouds may overlap and be determined as one cluster, causing the trajectory of one person to be lost. When PmTrack detects that the positions of two people are very close, a series of approaching processing including the determination of overlapping clusters is performed. This operation is conducted to segment overlapping clusters and associate them with multiple trajectories, and it has been proven to be very effective.

7.3.3 Trajectory Backtracking. If two people are already close when they appear in the radar FoV, we cannot determine the overlap based on their positions, as the radar will assume that only one person appears. We address this problem through two operations. The fine-grained clustering operation performs GMM clustering to segment

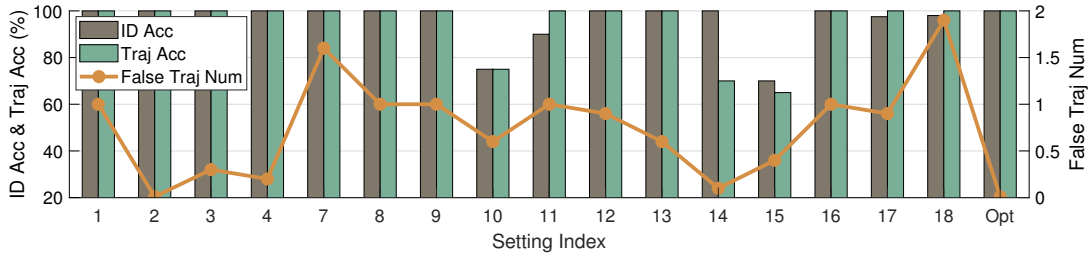


Fig. 36. Metrics after disabling the ghost suppression methods.

a cluster when the cluster width is too large. In addition, we perform trajectory backtracking on new targets that appear in the FoV to recover their missing history trajectories. In the experiments of setting 12, although the history of one person's trajectory is missing, the trajectory of the other person is complete. Since there are only two people in the space, it is still possible to achieve 100% identification accuracy by using the information from the other trajectory. However, in more complex scenarios with multiple targets, missing parts of trajectories can make identification more challenging. After trajectory backtracking, the new trajectories have all retrieved their historical segments, which proves that the method can ensure the integrity of the new trajectories.

7.3.4 Parallel Universe Competition. When two people approach and then separate, their trajectories may be associated with the wrong clusters, leading to trajectory errors. Settings 13 and 14 present two typical scenarios. In setting 13, two people briefly come close to each other before returning to their respective areas. Benefiting from the fine-grained clustering and approaching processing, the overlapping point clouds can be well segmented. In ten experiments, only one trajectory and identification error occurs. However, in setting 14, the two individuals cross their paths while walking, and it is difficult to accurately distinguish this situation by radar alone. As a result, half of the experiments occur trajectory errors, where the tracking system gives similar trajectories as setting 13. We propose a parallel universe competition mechanism that temporarily retains all trajectory-cluster association possibilities after trajectory separation and determines the universe that is consistent with the true situation via the average cost of each universe. This mechanism effectively prevents trajectory errors and ensures the accuracy of identification.

7.3.5 Waiting Zone Correction. In setting 15, the body-bounce ghost of participant B will pass by the position where participant A was standing. Since A is stationary while B is moving, and A is obscured by B, the reflection strength of A may be weaker than that of B's ghost. In five out of ten experiments, A's tracking trajectory is induced by B's ghost. The proposed solution is the waiting zone correction mechanism. When PmTrack detects that the trajectory may be induced by ghost clusters, it records the final position of the trajectory and observes whether the target reappears nearby within a certain period of time, so that the induced trajectory can be corrected. Our proposed stationary target enhancement further ensures the retrieval of the induced target.

7.3.6 Multipath Suppression. The body-bounce ghost from multipath propagation is a significant cause of various tracking errors. We propose a soft processing strategy that suppresses trajectories that use the cluster with ghost suspicion as an update object from entering the confirmed zone. Fig. 36 shows the results after disabling this mechanism. Although the specific degrees are related to human-radar distance, human-wall distance, trajectory interaction, etc., these experiments are all influenced by ghost clusters, resulting in trajectory and identification errors. As a comparison, our confirmation suppression mechanism prevents the occurrence of false trajectories and assists in high accuracy of tracking and identification.

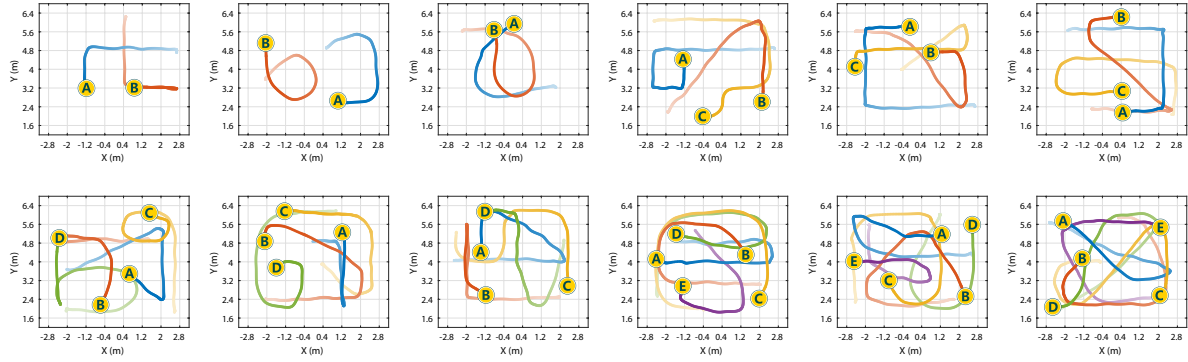


Fig. 37. Part of the results under free walking in the hall.

7.3.7 Gimbal Lock Correction. In the z-y-x rotation order, the gimbal lock problem occurs when the pitch of the IMU is close to $\pm 90^\circ$. At this time, the independence of yaw and roll is compromised, resulting in drastic changes and measurement errors of Euler angles. PmTrack can detect the occurrence of gimbal lock and utilize roll to correct yaw. To verify its effectiveness, we change the attitude of the IMU worn by the participants so that the y-axis is roughly parallel to the arm, and then conduct experiments under setting 8. As shown in Fig. 35, without correction, the similarity is only 69.54%, and the cost ratio reached 69.4%, indicating that the orientation measurement of the IMU is severely disturbed and can hardly be matched with the trajectory orientation, resulting in an identification accuracy of 80%. After correction, the similarity increases to 99.38%, and the cost ratio decreases to 13.23%, indicating that the correction successfully restored the discrimination between correct and incorrect pairs.

7.4 Performance under Free Walking

Previously the participants walk along preset trajectories, which may differ from actual walking patterns. In this section, the participants are asked to walk freely without being restricted by any predetermined trajectory. Experiments are conducted in both the hall and meeting room with 2~5 participants, and we perform 20 trials for each number of people. The upper time length of the orientations is 15s.

7.4.1 Free Walking in the Hall. We first conduct the evaluation in the hall. The collection time is 10s for 2 people and 15s for 3~5 people. Part of the results are plotted in Fig. 37, and Fig. 38 shows the identification accuracy and trajectory accuracy. When the number of people is no more than three, PmTrack can track and identify with 100% accuracy. As the number of people increases, trajectory errors begin to occur, leading to occasional identification errors. In the 5-person experiments, the system achieves an identification accuracy of 98% and a trajectory accuracy of 97%.

7.4.2 Free Walking in the Meeting Room. We then implement PmTrack in the meeting room, which is enclosed by walls in the radar FoV and contains items such as a table, chairs, bookshelves, and a monitor. The participants are free to sit in chairs. The collection time is 15s for 2~4 people and 30s for 5 people. Fig. 41 plots part of the tracking and identification results, and the accuracies are shown in Fig. 39. In this environment, the radar faces stronger multipath interference, and the accessible area of participants is reduced. Therefore, compared to the hall, the trajectory accuracy becomes lower, which then reduces the identification accuracy. Even so, PmTrack achieves

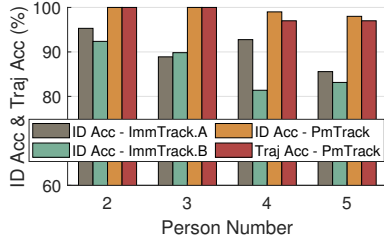


Fig. 38. Metrics under free walking in the hall.

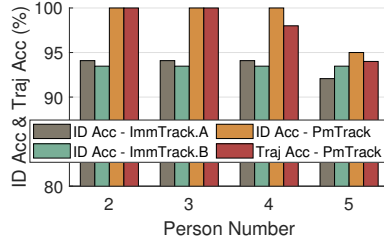


Fig. 39. Metrics under free walking in the meeting room.

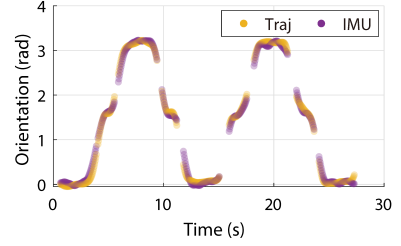


Fig. 40. Orientations under the square zig-zag path.

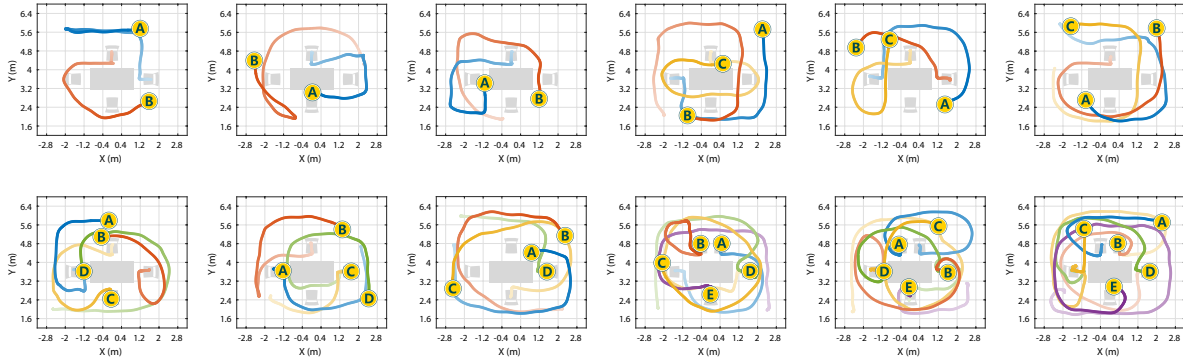


Fig. 41. Part of the results under free walking in the meeting room.

an identification accuracy of 95% and a trajectory accuracy of 94% in the 5-person experiments, demonstrating that our system can reliably operate in different application environments.

7.4.3 Comparison with State-of-the-Art. We compare the performance of PmTrack with ImmTrack [11], the most relevant and advanced research that utilizes IMU to identify radar targets. ImmTrack abstracts the radar-measured trajectory and IMU trajectory as feature images, and then associates them using a Siamese network. PmTrack's identification accuracy in the hall and meeting room is respectively compared with that of ImmTrack in sport hall and lab space. The metrics of ImmTrack are also shown in Fig. 38 and Fig. 39, where the suffixes ".A" and ".B" represent its single-window and multi-window modes, respectively.

It can be seen that PmTrack performs better in both environments. We attribute this result to the accuracy and reliability of using orientation as the matching feature. We perform the same matching feature demonstration experiment as ImmTrack, where a participant walks along a square zig-zag path. Fig. 40 shows the orientations of trajectory and IMU, and we can see that the two are highly consistent. In fact, in all of the experiments, the orientation similarity of correct pairs is almost all higher than 99%, while the similarity of incorrect pairs decreases sharply. As a result, as long as the radar-measured trajectory is correct, PmTrack can easily and accurately achieve matching without being affected by the cumulative error of the accelerometer.

8 RELATED WORK

8.1 Device-free Tracking

Compared with tracking technologies based on signals such as WiFi [47, 67] and acoustics [3], mmWave radar has advantages in deployment complexity and localization accuracy. CornerRadar [73] utilizes multipath signals to locate objects around corners. For indoor multi-person tracking, with the improvement of the angular resolution of single-device radar, recent works tend to use RA point clouds in the target detection stage [8, 9, 40, 79]. In trajectory management, TI [56] proposes a trajectory zone scheme and considers stationary targets. We conduct a more comprehensive design of the tracking system so that it can cope with various practical problems. Some researchers [64] have noticed the point cloud overlap of adjacent people and propose to introduce methods such as GMM for re-clustering, whereas only a few studies [4, 45] are concerned with identity re-association after the separation of overlapping people. The solution to this problem is to perform human identification, which will be discussed later. We propose a parallel universe competition mechanism for trajectory separation, which fits well with IMU-assisted identification and effectively solves the problem of trajectory-cluster association errors.

Regarding the ghost reflection, the most intuitive solution is to restrict the point cloud coordinate according to the building structure [45], but this cannot deal with body-bounce ghosts. Methods based on synthetic aperture radar imaging require a larger antenna array [19, 34]. The classification of targets using RD information [16] cannot cope with the overlap of a real person and a ghost in the RD map. Complementing multiple radar viewpoints [9, 18] has a very good effect because the ghost targets observed from different positions vary greatly, but it increases the cost of purchasing, deploying, and computing. In summary, existing methods cannot accurately recognize body-bounce ghosts in mmWave tracking. Our work provides moderate flexibility to this challenge with soft decision-making methods, minimizing the impact of ghost reflection on multi-person tracking.

8.2 RF-based Human Identification

Some studies [4, 5] utilize computer vision to identify radar targets, which may bring concerns about privacy leakage. As an alternative, researchers often utilize information that reflects individual differences. Among them, gait information obtained through micro-Doppler spectra [39, 42, 45] or stride measurements [72] is the most common feature. In addition, the body 3D point cloud [7, 79] and behavior pattern [22] are also a considered factor. Some identity recognition methods for stationary targets take advantage of differences in people's facial contours [68], voiceprints [25], and respiratory features [62]. The advantage of above schemes is that they are completely device-free and can be done by radar alone. However, the collection of training data is inevitable, and the diversity and variability of application scenarios also impact the feature matching. Moreover, the distinctiveness of signal features decreases as the number of people increases.

One approach to address the limitations of individual difference-based schemes is to use additional sensors as identity indicators. The dependence of passive RFID [26, 35] on LoS path damages its interrogation performance and signal feature availability in free walking scenarios. Semi-active RFID tags [46], WiFi tags [58], Bluetooth tags [81], and UWB tags [49], etc. require specialized transceiving equipment or structures. mmWave Van Atta arrays [2, 38, 53] do not require additional equipment and can be passive [41], but so far have not been miniaturized to the extent that they can be easily carried. In contrast, IMUs exist in portable devices such as smartwatches and smartphones with negligible volume and power consumption, and their data uploading is based on existing wireless networks, making them ideally suited for assisting the radar target identification in a lightweight way. ImmTrack [11] also utilizes IMU to identify radar targets, but its dependence on trajectory, which is susceptible to serious interference from cumulative errors in the IMU system, limits its matching performance. PmTrack utilizes the orientation that is accurate and reliable as the matching feature, achieving superior matching performance.

8.3 mmWave-based Human Sensing

mmWave radar significantly enhances the performance of RF-based human sensing. In addition to tracking, many types of applications have sprung up to enable intelligent sensing and monitoring. Based on the measurement of chest vibrations, various methods have been proposed to monitor vital signs such as respiratory rate, heart rate [6, 10, 69], and even subtle heart activities [20, 23]. The analysis of micro-Doppler spectra and spatial spectra endows radar with the ability to observe human movements at different scales, including hand gestures [15, 21, 36, 63, 66], arm movements [28, 30, 33, 44, 50, 65], and overall activities [13, 27, 48, 61, 75, 78]. The description of human contours is a promising topic [1, 60], which includes skeleton estimation [24, 54, 77] and mesh reconstruction [14, 70, 71]. Benefiting from the displacement sensitivity of mmWave radar, vibrations of the vocal cords [25, 43, 59, 74] or surrounding objects [76] can be converted into acoustic spectrograms, enabling microphone-free sound acquisition. The powerful sensing capabilities also bring greater risks of privacy leakage, and research on privacy protection of human activities is emerging [31, 52]. These technologies have broad prospects in home, security, and medical fields. Our research can provide identity directionality for them to further improve the sensing quality and diversity.

9 CONCLUSION & DISCUSSION

This paper presented the PmTrack system, which introduces IMUs as identity indicators for empowering mmWave human tracking and sensing technologies with accurate, lightweight, and highly feasible identity recognition capability. By using orientation as the matching feature, PmTrack addresses the data heterogeneity between radar and IMU, and meanwhile gets rid of the cumulative error of the accelerometer. To overcome practical difficulties caused by the three major challenges of weak reflection, point cloud overlap, and body-bounce ghost in multi-person tracking, this paper proposed a variety of optimization methods for detection enhancement, interference suppression, continuity maintenance, and trajectory correction to ensure the accuracy of the trajectories. We also proposed an orientation correction method to mitigate the effect of IMU gimbal lock error. Experimental results reveal that PmTrack achieves an identification accuracy of 98% and 95% with five people in the hall and meeting room, respectively. PmTrack utilizes the advantages of both the mmWave system and IMU system reasonably and efficiently, thereby expanding the development and application of sensing technology.

It should be emphasized that PmTrack's dependence on IMU data requires users to open relevant interfaces on their mobile devices, which potentially compromises their activity privacy. Our opinion is as follows. The data interface only needs to be opened when enabling sensing system's identification function, and at this point, users' activity information is already being detected by the radar, implying that the application scenario is not sensitive to activity privacy. Moreover, PmTrack is a plug-and-play algorithm based on common mmWave radar systems, so its enabling does not impose additional burden on users, nor does its disabling cause losses to users. Although PmTrack inevitably brings privacy risks, it does create opportunities for users to enjoy personalized tracking and sensing services, and users can decide whether they are willing to bear this risk.

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